

Week 2 Practical

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MySQL 8.0 Command Line Cli  ×  +  ∨

Enter password: *****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.0.22 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE bookflow_db;
ERROR 1007 (HY000): Can't create database 'bookflow_db'; database exists
mysql> Use bookflow_db;
Database changed
mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| books                  |
| members               |
+-----+
2 rows in set (0.03 sec)

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title       | isbn       | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist | 9780061122415 | 1988 |
| 2       | Clean Code   | 9780132350884 | 2008 |
| 3       | Atomic Habits | 9780735211292 | 2018 |
+-----+-----+-----+-----+
3 rows in set (0.04 sec)
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MySQL 8.0 Command Line Cli  x  +  v
mysql> SELECT * FROM Members;
+-----+-----+-----+
| member_id | full_name | email |
+-----+-----+-----+
| 1 | Anil Kumar | anil.kumar@example.com |
| 2 | Priya Sharma | priya.sharma@example.com |
| 3 | Ravi Verma | ravi.verma@example.com |
+-----+-----+-----+
3 rows in set (0.02 sec)

mysql> CREATE TABLE Loans (loan_id INT PRIMARY KEY, member_id INT, book_id INT, loan_date DATE, FOREIGN KEY (member_id) REFERENCES Members(member_id), FOREIGN KEY (book_id) REFERENCES Books(book_id));
Query OK, 0 rows affected (0.08 sec)

mysql> SHOW TABLES;
+-----+
| Tables_in_bookflow_db |
+-----+
| books |
| loans |
| members |
+-----+
3 rows in set (0.00 sec)

mysql> INSERT INTO Loans (loan_id, member_id, book_id, loan_date) VALUES(1, 1, 1, '2026-08-01'),(2, 2, 2, '2026-08-02'),(3, 3, 3, '2026-08-03'),(4, 1, 2, '2026-08-04'),(5, 2, 1, '2026-08-05'),(6, 3, 2, '2026-08-06'),(7, 1, 3, '2026-08-07'),(8, 2, 3, '2026-08-08'),(9, 3, 1, '2026-08-09'),(10, 1, 1, '2026-08-10');
Query OK, 10 rows affected (0.02 sec)
Records: 10 Duplicates: 0 Warnings: 0

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MySQL 8.0 Command Line Cli  x  +  v
mysql> SELECT * FROM Loans;
+-----+-----+-----+-----+
| loan_id | member_id | book_id | loan_date |
+-----+-----+-----+-----+
| 1 | 1 | 1 | 2026-08-01 |
| 2 | 2 | 2 | 2026-08-02 |
| 3 | 3 | 3 | 2026-08-03 |
| 4 | 1 | 2 | 2026-08-04 |
| 5 | 2 | 1 | 2026-08-05 |
| 6 | 3 | 2 | 2026-08-06 |
| 7 | 1 | 3 | 2026-08-07 |
| 8 | 2 | 3 | 2026-08-08 |
| 9 | 3 | 1 | 2026-08-09 |
| 10 | 1 | 1 | 2026-08-10 |
+-----+-----+-----+-----+
10 rows in set (0.00 sec)

mysql> SELECT m.full_name AS Member_Name, b.title AS Book_Title FROM Loans l INNER JOIN Members m ON l.member_id = m.member_id INNER JOIN Books b ON l.book_id = b.book_id;
+-----+-----+
| Member_Name | Book_Title |
+-----+-----+
| Anil Kumar | The Alchemist |
| Anil Kumar | Clean Code |
| Anil Kumar | Atomic Habits |
| Anil Kumar | The Alchemist |
| Priya Sharma | Clean Code |
| Priya Sharma | The Alchemist |
| Priya Sharma | Atomic Habits |
| Ravi Verma | Atomic Habits |
| Ravi Verma | Clean Code |
| Ravi Verma | The Alchemist |
+-----+-----+
10 rows in set (0.00 sec)

```

```
MySQL 8.0 Command Line Cli  x  +  v

mysql> SELECT published_year, COUNT(book_id) AS Total_Books FROM Books GROUP BY published_year ORDER BY published_year;
+-----+-----+
| published_year | Total_Books |
+-----+-----+
| 1988          | 1          |
| 2008          | 1          |
| 2018          | 1          |
+-----+-----+
3 rows in set (0.01 sec)

mysql> CREATE TABLE Donation_History ( donation_id INT PRIMARY KEY, book_id INT, donor_name VARCHAR(100), donation_date DATE, FOREIGN
KEY (book_id) REFERENCES Books(book_id));
Query OK, 0 rows affected (0.05 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> INSERT INTO Books (book_id, title, isbn, published_year) VALUES (4, 'The Power of Habit', '9780812981605', 2012);
Query OK, 1 row affected (0.00 sec)

mysql> INSERT INTO Donation_History (donation_id, book_id, donor_name, donation_date) VALUES (1, 4, 'Raj Kumar', CURDATE());
Query OK, 1 row affected (0.01 sec)

mysql> COMMIT;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Books;
+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist   | 9780061122415 | 1988           |
| 2       | Clean Code      | 9780132350884 | 2008           |
| 3       | Atomic Habits   | 9780735211292 | 2018           |
| 4       | The Power of Habit | 9780812981605 | 2012           |
+-----+-----+-----+-----+
4 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Donation_History;
+-----+-----+-----+-----+
| donation_id | book_id | donor_name | donation_date |
+-----+-----+-----+-----+
| 1           | 4       | Raj Kumar  | 2026-08-12    |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> CREATE INDEX idx_books_isbn ON Books(isbn);
Query OK, 0 rows affected (0.06 sec)
Records: 0 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Books WHERE isbn = '9780061122415';
+-----+-----+-----+-----+
| book_id | title           | isbn           | published_year |
+-----+-----+-----+-----+
| 1       | The Alchemist   | 9780061122415 | 1988           |
+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql> |
```